

```

# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

# -----
# MOUNT GOOGLE DRIVE (IMPORTANT)
# -----
from google.colab import drive
drive.mount('/content/drive')

# -----
# LOAD DATASET
# -----
data = pd.read_csv('/content/drive/MyDrive/svm_dataset.csv')

# Features and labels
X = data[['X1', 'X2']]
y = data['Y']

# -----
# TRAIN TEST SPLIT (70-30)
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# -----
# TRAIN SVM MODEL
# -----
model = SVC(kernel='linear')
model.fit(X_train, y_train)

# -----
# MODEL ACCURACY
# -----
accuracy = model.score(X_test, y_test)
print("Accuracy:", accuracy)

# -----
# PLOT DATA
# -----
plt.figure()
plt.scatter(X['X1'], X['X2'], c=y)

# -----
# DECISION BOUNDARY
# -----
ax = plt.gca()
xlim = ax.get_xlim()
ylim = ax.get_ylim()

# Create grid
xx = np.linspace(xlim[0], xlim[1], 50)
yy = np.linspace(ylim[0], ylim[1], 50)
YY, XX = np.meshgrid(yy, xx)

xy = np.vstack([XX.ravel(), YY.ravel()]).T
Z = model.decision_function(xy)
Z = Z.reshape(XX.shape)

# Plot boundary + margins
ax.contour(XX, YY, Z,
           levels=[-1, 0, 1],
           linestyles=['--', '-', '--'])

# -----
# SUPPORT VECTORS
# -----
plt.scatter(model.support_vectors_[0],
            model.support_vectors_[1],
            s=100, facecolors='none', edgecolors='black')

```

```
# Labels
plt.xlabel("X1")
plt.ylabel("X2")
plt.title("SVM Decision Boundary")

plt.show()
```

Mounted at /content/drive

Accuracy: 1.0

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but warnings.warn(

